

Lab 7: SD Card & Web Server

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Objective

The objective of this lab was to create a simple web server using the ESP32 that serves an HTML webpage stored on an SD card. The ESP32 connects to a Wi-Fi network and hosts a web server that responds to browser requests by sending the stored HTML file. This lab demonstrates the integration of wireless communication, file systems, and embedded web hosting.

Materials

- ESP32 Development Board
- Micro SD Card Module
- SD Card (with HTML file)

Background and Theory

Wi-Fi Communication (Station Mode)

The ESP32 has a Wi-Fi module built in that lets it connect to a wireless network. In this lab, the ESP32 is in Station (STA) mode, which means it connects to a router that is already set up using an SSID and password. When you connect, the ESP32 gets an IP address that you can use to access it over the network.

Web Server Concept

A web server is a system that listens for incoming HTTP requests from a client (such as a web browser) and responds with data, typically in the form of an HTML webpage. When a user enters the ESP32's IP address into a browser:

1. The browser sends an HTTP request
2. The ESP32 receives the request
3. The ESP32 reads the HTML file from the SD card
4. The HTML is sent back and rendered in the browser

SD Card File System

The ESP32 uses the SD card as external storage. The HTML file is stored on the SD card instead of being hardcoded into the program. It is read dynamically when it is needed. This makes it easier to change the webpage without having to change the code.

Procedure

1. Created a basic HTML file (index.html) for the webpage.
2. Saved the HTML file onto the SD card.
3. Inserted the SD card into the SD card module.
4. Connected the SD card module to the ESP32 using the specified GPIO pins.
5. Opened Arduino IDE and included required libraries for Wi-Fi and SD card communication.
6. Programmed the ESP32 to connect to a Wi-Fi network using SSID and password.
7. Initialized the SD card using SPI communication.
8. Created a web server using the ESP32.
9. Programmed the server to read the index.html file from the SD card when a client connects.
10. Uploaded the code to the ESP32.

11. Opened the Serial Monitor to obtain the assigned IP address.
12. Entered the IP address into a web browser to view the webpage.

Code:

```
#include <WiFi.h>
#include <WebServer.h>
#include <SD_MMC.h>

const char* ssid = "JSU_Wireless";

WebServer server(80);

void handleRoot() {
    File file = SD_MMC.open("/index.html", FILE_READ);

    if (!file) {
        server.send(404, "text/plain", "index.html not found on SD card!");
        return;
    }

    String html = "";
    while (file.available()) {
        html += (char)file.read();
    }
    file.close();

    server.send(200, "text/html", html);
}

void setup() {
    Serial.begin(115200);
    delay(10);

    if (!SD_MMC.begin("/sdcard", true)) {
        Serial.println("SD card failed! Is it inserted?");
        return;
    }
    Serial.println("SD card ready!");

    // Connect to WiFi
    WiFi.begin(ssid);
```

```

Serial.print("Connecting to WiFi");
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("\nWiFi Connected!");
Serial.print("Type this IP in your browser: ");
Serial.println(WiFi.localIP());

server.on("/", handleRoot);
server.begin();
Serial.println("Web server is running!");
}

void loop() {
    server.handleClient();
}

```

HTML Code (index.html):

```

board_manager:
  additional_urls: []
network:
  connection_timeout: 1000s
directories:
  builtin:
    libraries: C:\Users\stanl\AppData\Local\Arduino15\libraries
    data: c:\Users\stanl\AppData\Local\Arduino15
    user: c:\Users\stanl\OneDrive\Documents\Arduino
locale: en

```

Results

The ESP32 successfully connected to the Wi-Fi network and displayed its assigned IP address in the Serial Monitor. After entering the IP address into a web browser, the HTML page stored on the SD card was successfully loaded and displayed.

The server responded consistently to repeated requests, demonstrating stable communication between the ESP32 and the client device. The SD card was properly initialized, and the file reading process worked as expected without errors.

Conclusion

This lab successfully demonstrated how to create a web server using the ESP32 and serve a webpage stored on an SD card. The ESP32 operated in Station mode to connect to a Wi-Fi network and used its assigned IP address to communicate with a web browser.